Basic MOS Device Physics.

In studying the design of integrated circuit one of two approaches can be taken:

(1) begin with quantum mechanics and understand solid-state physics;

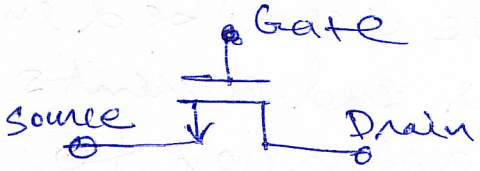
(2) treat each semiconductor device as a black box whose behavior is described in terms of its terminal voltages and currents. design circuits with little attention to the internal operation of the device.

In this chapter, we study the physics of MOSFETs at an elementary level, covering the minimum that is necessary for basic analog design. We begin our study with the structure of MOS transistors and derive their I/V characteristics. Next we describe second-order effects such as body effect, channel-length modulation, and subthreshold condition. We then identify the parasitic capacitances of MOSFETs, derive a small-signal model.

General Considerations:

1. MOSFET as a switch

Shown in Fig. is the symbol for an n-type MOSFET, revealing three terminals: gate (G), source (S), and drain (D). The latter two are interchangeable because the device is symmetric

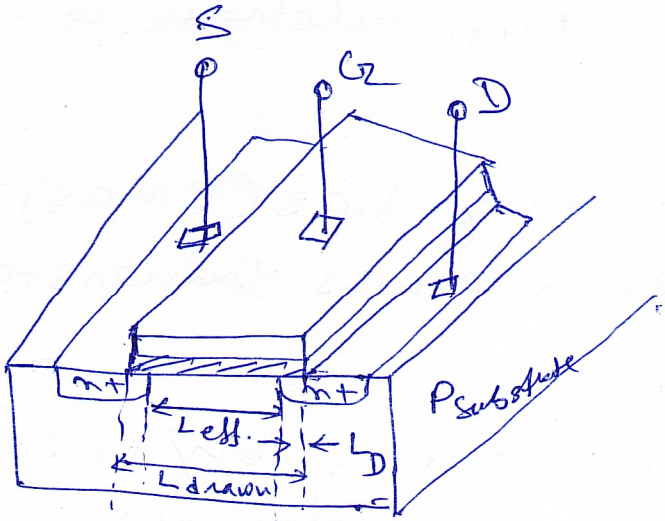


when operating as a switch, the transistor connects the source and the drain together if the gate voltage, V_G, "high" and isolates the source and the drain if V_G is "low".

Even with this simplified view, we must answer several questions. For what value of V_G does the device turn on? In other words, what's the "threshold" voltage? What is the resistance bet" S and D when the device is on (or off)? How does this resistance depend on the terminal voltages? Can we always model the path bet" S and D by a simple linear resistor? What limits the speed of the device?

2. MOSFET Structure:

Fig. shows a simplified structure of an n-type MOS (NMOS) device. Fabricated on a p-type substrate (also called the "bulk" or the body"). The device consists of two heavily doped n regions forming the source and drain terminals, a heavily-doped (conductive) piece of polysilicon (often simply called "Poly") operating as the gate, and a thin layer of silicon dioxide (SiO_2) insulating the gate from the substrate. The useful action of the device occurs in the substrate region under the gate oxide. Note that the structure is symmetric with respect to S and D.



The dimension of the gate along the source-drain path is called the length, L, and that perpendicular to the length is called the width, W. Since during fabrication the S/D junctions "side diffuse," the actual distance bet" the source and the drain is slightly less than L. To avoid confusion, we write, $L_{eff} = L_{drawn} - 2L_D$, where L_{eff} is the effective length,

L_{drawn} is the total length, and L_D is the amount of side diffusion.

We have thus far ignored the substrate on which the device is fabricated. In reality, the substrate potential greatly influences the device characteristics. That is, the MOSFET is a four-terminal device. Since in typical NOS operation the S/D junction diodes must be reverse-biased.

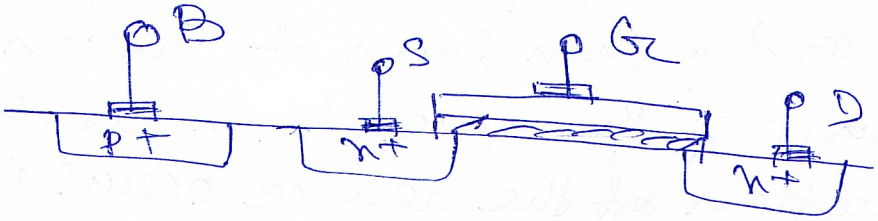
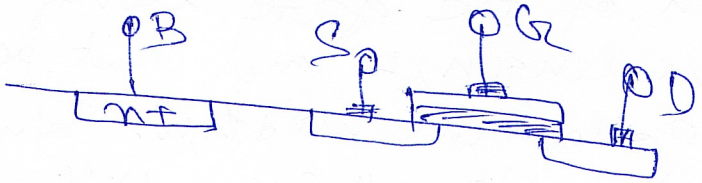


Fig. Substrate connections.

In complementary MOS (CMOS) technologies, both NMOS and PMOS transistors are available. From a simplistic view point, the PMOS device is obtained by negating all of the doping types (including the substrate).



but in practice, NMOS and PMOS devices must be fabricated on the same wafer, i.e the same substrate. For this reason, one device type can be placed in a "local substrate," usually called a "well." In most of today's cmos processes, the PMOS device is fabricated in an n-well. Note that the n-well must be connected to a potential such that the S/D junction-diodes of the PMOS transistor remain reverse-biased under all conditions. In most circuits, the n-well is tied to the most positive supply voltage. We sometimes call NMOS and PMOS devices "NFETs" and "PFETs," respectively.

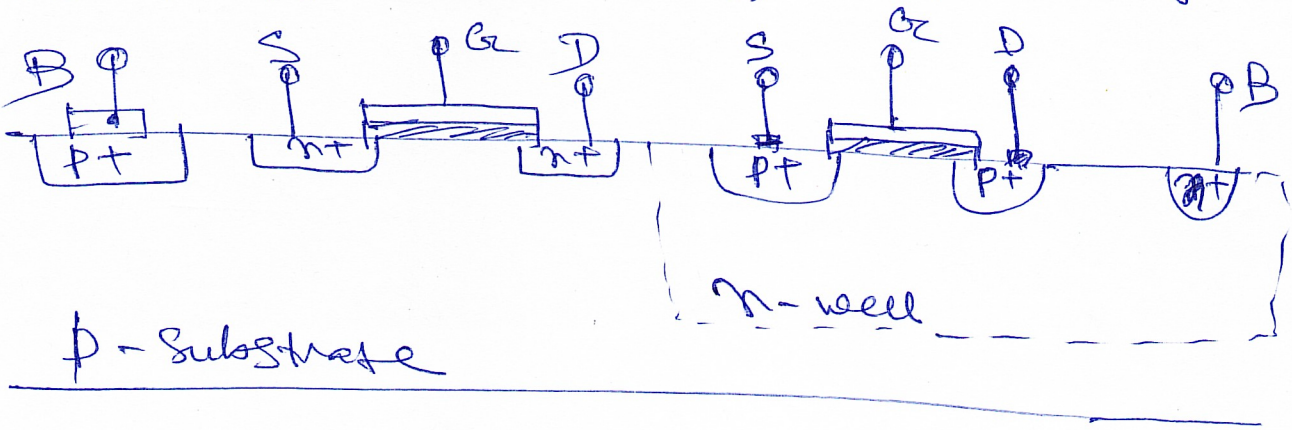


Fig. PMOS inside an n-well

Mos Symbols:

The circuit symbol used to present NMOS and PMOS transistors are shown in Fig.

